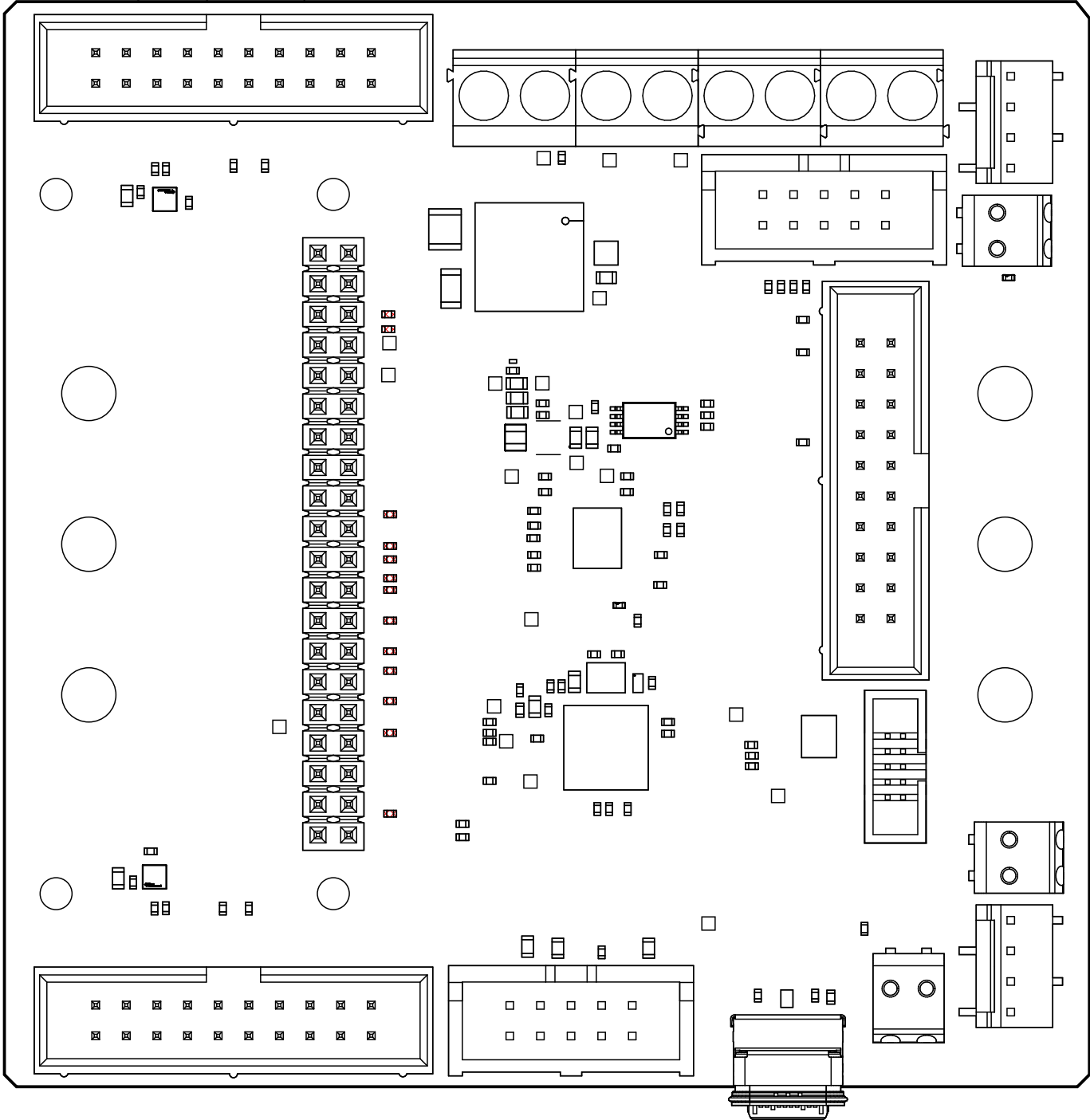
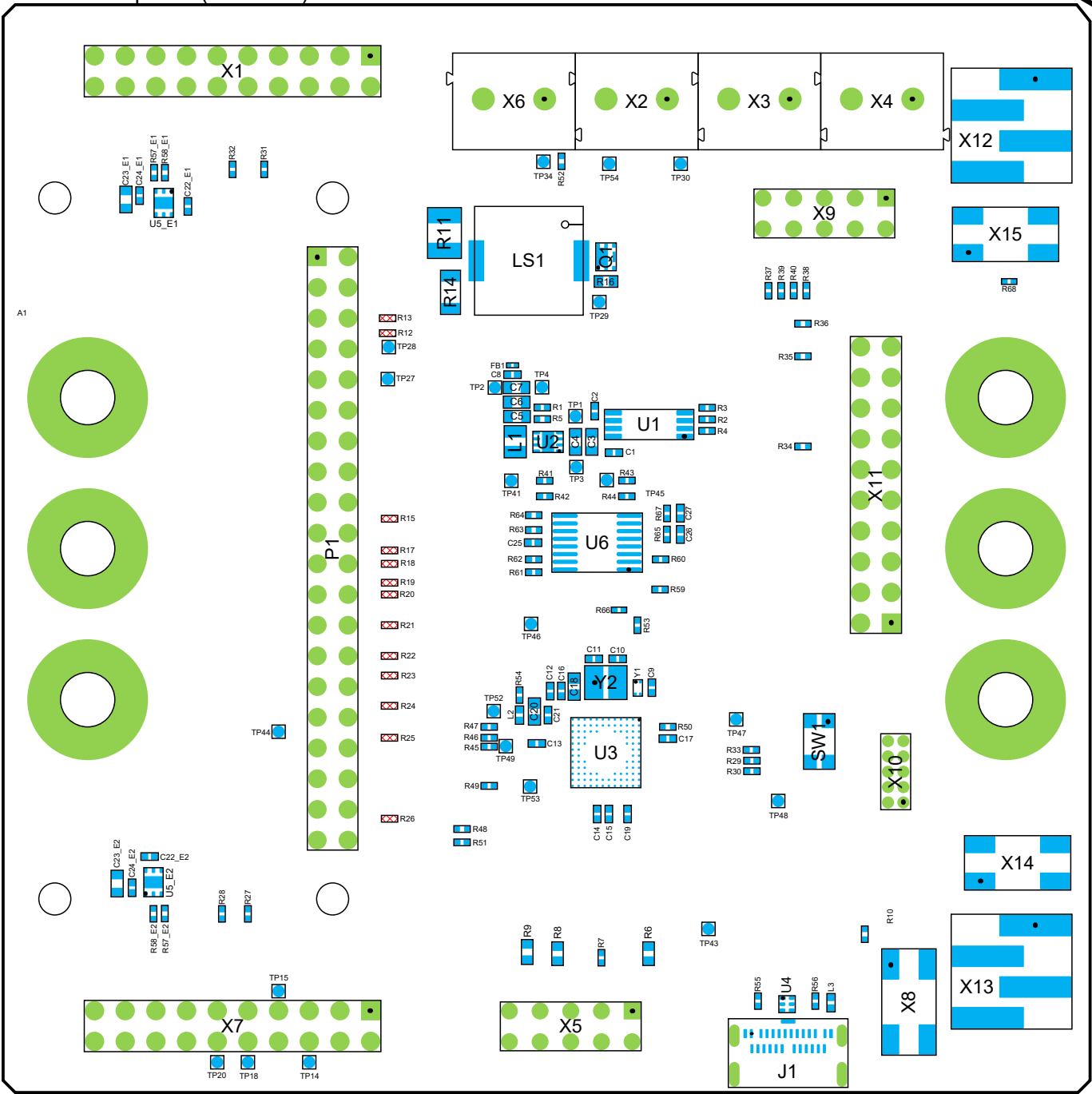


Assembly documentation:

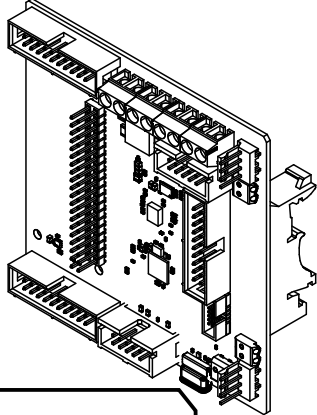
View from Top side (Scale 2:1)



View from Top side (Scale 2:1)



View from Top side (Scale 1:2)



- Notes:
- 1. Pin 1 marked with dot
 - 2. Cathode marked with K

			Project: STB - Control Board	
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Bill Of Materials									
Line #	Quantity	Designator	Name	Value	Tolerance	Power	Footprint	Manufacturer	Manufacturer Part Number
	17	C1, C2, C8, C9, C12, C13, C14, C15, C16, C17, C19, C21, C22_E1, C22_E2, C24_E1, C24_E2, C25	100nF	100nF			CAPC1050X55X23LL05T13	Samsung Electro-Mechanics	CL05B104KO5NNNC
	7	C3, C4, C5, C6, C7, C23_E1, C23_E2	10uF	10uF	±10%		CAPC0603(1608)100_L	Samsung	CL10A106MP8NNNC
	2	C10, C11	16pF	16pF			CAPC1005X55X25LL05T10	Murata	GRM1555C1H160JA01D
	2	C18, C20	1uF	1uF			CAPC1608X90X35LL15T15	Kemet	C0603C105K4RACTU
	2	C26, C27	100pF	100pF			CAPC1005X56X25LL10T15	AVX Interconnect / Elco	04023A101KAT2A
	1	FB1	80mR		25%		MURA-BLM03-CHIP-2_V	Murata	BLM03AX800SN1D
	1	J1	2305018-2				2305018-2	TE Connectivity	2305018-2
	1	L1	DFE201612E-2R2M=P2	2.2uH	±20%		INDC2016X10L	Murata Electronics	DFE201612E-2R2M=P2
	2	L2, L3	LI0402E300R-10				INDC1005X07L	Laird-Signal Integrity Products	LI0402E300R-10
	1	LS1	PKMCS0909E4000-R1				MURA- PKMCS0909E4000-2_V	Murata	PKMCS0909E4000-R1
	2	M1, M2	1202713				1202713	Phoenix Contact	1202713
	1	P1	TSW-120-24-L-D				SMTC-TSW-120-24-X-D	Samtec	TSW-120-24-L-D
	1	Q1	SIA440DJ-T1-GE3				SC70-6L_65P210X80_HS-7L	Vishay Siliconix	SIA440DJ-T1-GE3
	2	R1, R2	470k	470 kOhm	1%		RESC1005X40X25LL05T10	Yageo / Phycomp	RC0402FR-07470KL
	1	R3	1k	1 kOhm	1%		RESC1005X40X25LL05T10	Yageo / Phycomp	RC0402FR-071KL
	8	R4, R7, R41, R42, R43, R44, R51, R53	100k	100 kOhm	1%		RESC1005X40X25LL05T10	Yageo / Phycomp	RC0402FR-07100KL
	1	R5	150k	150 kOhm	1%		RESC1005X40X25LL05T05	Yageo	RC0402FR-07150KL
	2	R6, R16	56R	56 Ohm	1%		RESC1609X50X30LL10T20	Vishay Dale	CRCW060356R0FKEA
	1	R8	30R	30 Ohm	1%		RESC1609X50X30LL10T20	Vishay Dale	CRCW060330R0FKEA
	1	R9	27R	27 Ohm	1%		RESC1608X55X30LL15T15	Panasonic	ERJ-3EKF27R0V
	7	R10, R34, R35, R36, R50, R63, R64	10k	10 kOhm	1%		RESC1005X40X25LL05T05	Panasonic	ERJ-2RKF1002X
	1	R11	1R	1 Ohm	1%		RESC3225X70X50NL20T20	Panasonic	ERJ14BQF1R0U
	1	R14	1k	1 kOhm	1%		RESC3216X70X50LL05T20	Panasonic	ERJ-8ENF1001V
	10	R27, R28, R31, R32, R37, R38, R59, R60, R61, R62	4k7	4.7 kOhm	1%		RESC1005X40X25LL05T10	Yageo / Phycomp	RC0402FR-074K7L
	10	R29, R30, R33, R39, R40, R45, R46, R47, R49, R54	0R	0R			RESC1005X40X25LL05T10	Yageo / Phycomp	RC0402JR-070RL
	5	R48, R57_E1, R57_E2, R58_E1, R58_E2	390k	390 kOhm	1%		RESC1005X40X25LL05T05	Panasonic	ERJ-2RKF3903X
	1	R52	820k	820 kOhm	1%		RESC1005X40X25LL05T05	Vishay Dale	CRCW0402820KFKED
	2	R55, R56	5k1	5.1 kOhm	1%		RESC1005X40X25LL05T10	Yageo / Phycomp	RC0402FR-075K1L
	2	R65, R67	1k6	1.6 kOhm	1%		RESC1005X40X25LL05T10	Yageo / Phycomp	RC0402FR-071K6L
	2	R66, R68	NCP15XH103F03RC	10 kOhm	1%		FP-NCP15-NCU15-0_05-IPC_C	Murata	NCP15XH103F03RC
	1	SW1	EVP-AA002K				PNSC-EVPAA_V	Panasonic	EVP-AA002K
	1	U1	TPS2111APW				PW0008A_N	Texas Instruments	TPS2111APW
	1	U2	TPS62160QDSGRQ1				FP-DSG0008A-IPC_C	Texas Instruments	TPS62160QDSGRQ1
	1	U3	STM32L452VEI6				UFBGA100	STMicroelectronics	STM32L452VEI6
	1	U4	TPD4S009DRYR				DRY0006A	Texas Instruments	TPD4S009DRYR
	2	U5_E1, U5_E2	TPS25200DRV				FP-DRV0006A-MFG	Texas Instruments	TPS25200DRV
	1	U6	MAX6615AEE+T				MAXM-E16+1_M	Maxim	MAX6615AEE+T
	3	X1, X7, X11	61202021621				61202021621	Würth Elektronik	61202021621
	4	X2, X3, X4, X6	1760490000				PM_5_08_90_02	Weidmueller	1760490000
	2	X5, X9	5103309-1				TECO-5103309-1_V	TE Connectivity	5103309-1
	3	X8, X14, X15	2059-302/998-403				2059-302/998-403	WAGO	2059-302/998-403
	1	X10	3220-10-0100-00				3220-10-0100-00	CNC Tech	3220-10-0100-00
	2	X12, X13	3-647168-4				3-647168-4	TE Connectivity	3-647166-4
	1	Y1	SiT1532AI-J4-DCC-32.768E				4-UFBGA	SiTime	SiT1532AI-J4-DCC-32.768E
	1	Y2	8MHz				NDK-NX3225GD-8MHZ-STD-CRA-3_V	NDK	NX3225GD-8MHZ-STD-CRA-3

ETH

zürich

Drawing number:

Rev: B

Date: 23.11.2022

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A3

Project:

STB - Control Board

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A	B	C	D	E	F	G	H
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